

ACRME Region Selection WHAT-IF Tool — User Guide

Workbook: `acrme_region_selection_whatif_acrme_regions.xlsx`

Baseline: Azure Capacity & Quota Management Consolidated Requirements Baseline v2.4

Status: Mockup / Planning Tool — derived planning figures only; not connected to live Azure state

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1. Purpose and Scope

This workbook implements a **weighted capacity placement model** for the ACRME (Automated Customer-Region Management Engine) programme. Given a customer's geography, SKU, and VM count, it evaluates all 14 ACRME in-scope regions and recommends:

- **Prod region** — the production placement
- **CVAL region** — the Customer Validation / non-prod placement
- **DR region** — the disaster recovery placement

The tool is a **planning mockup**, not a live API. Capacity figures are derived from real usage data (distributed onto the baseline v2.4 region catalogue per the builder script) and are refreshed by re-running `build_acrme_whatif_acrme_regions.py`. All cell formulas respond instantly to edits in the yellow input cells and the Capacity_Usage data rows — no macro or recalculation button is required.

Scope of This Guide

This guide covers:

- How to update capacity and quota data (with worked examples)
- How to model multi-SKU customer deployments
- A full explanation of every formula in the workbook, traced to the actual cell references

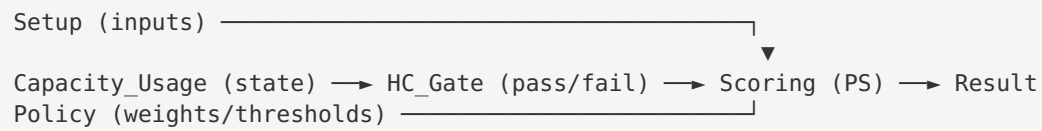
Out of scope: modifying the 14-region catalogue (CATALOGUE list in the builder), changing geographic distribution weights, or rerunning the builder script — those are engineering activities requiring a rebuild.

2. Workbook Architecture

The workbook contains 10 sheets in processing order:

Order	Sheet	Role	User-editable
1	How_To_Use	Navigation guide and colour legend	No
2	Usage_Source	Raw source data imported from SKU_USage.xlsx and sku_usage_prod_v1.xlsx	No
3	Setup	Scenario inputs: geography, SKU, VM count, customer ID	Yellow cells only
4	Policy	Weight parameters, headroom thresholds, total_customers mode	Yellow cells only
5	Capacity_Usage	Capacity and quota state for all 14 ACRME regions	Specific columns (see §5.1)
6	HC_Gate	Hard-constraint eligibility checks for each region	No — all formulas
7	Scoring_Prod	Weighted placement scoring for the Prod environment	No — all formulas
8	Scoring_CVAL	Weighted placement scoring for the CVAL environment	No — all formulas
9	Scoring_DR	Weighted placement scoring for the DR environment	No — all formulas
10	Result	Recommended Prod/CVAL/DR regions and ranked candidate table	No — all formulas

Data Flow (text diagram)



Every computed cell is a live Excel formula — edits to Setup, Policy, or Capacity_Usage propagate automatically throughout the chain.

The 14 ACRME Regions (baseline v2.4, Section 6)

Row	Region Name	Region ID	Geo-graphy	AZ	Model	Class
4	West US 3	westus3	US	3	3-region	Standard
5	Central US	centralus	US	3	3-region	Standard
6	Canada Central	canada-central	US	3	3-region	Standard
7	East US 2	eastus2	US	3	3-region	Standard
8	Switzerland North	switzerlandnorth	EU	3	2-region	Standard
9	Sweden Central	sweden-central	EU	3	2-region	Standard
10	North Europe	northeurope	EU	3	2-region	Standard
11	West Europe	west-europe	EU	3	2-region	Standard
12	Australia East	australiaeast	AU	3	2-region	Standard
13	Australia Southeast	australia-southeast	AU	2	2-region	Standard
14	East Asia	eastasia	AP	3	2-region	Standard
15	Southeast Asia	southeast-asia	AP	3	2-region	Standard
16	Saudi Arabia Central	saudicentral	ME	3	cross-geo	Restricted
17	UAE North	uaenorth	ME	3	cross-geo	Restricted

AU = Australia · AP = Asia Pacific · ME = Middle East

Restricted = not auto-selected by the engine; requires the Prod region to be explicitly named in Setup!B8.

3. Reading the Result Sheet

Open **Result** (the last sheet) to see the recommended placements for the current scenario.

Primary Placements (rows 9-11)

Cell	Content
B9	Recommended Prod region (or NONE ELI-GIBLE)
C9	Prod placement score (0.00-1.00)
B10	Recommended CVAL region (or NONE ELI-GIBLE)
C10	CVAL placement score
B11	Recommended DR region (or NONE ELI-GIBLE)
C11	DR placement score

Readiness State (B13)

Value	Meaning
READY	All three placements found; deployment can proceed
QUOTA_DEFICIT / NEEDS EXPLICIT REGION (Prod)	No Prod-eligible region — check quota data in Capacity_Usage and/or name the Prod region explicitly in Setup!B8
CAPACITY_UNAVAILABLE (CVAL)	No CVAL-eligible region in the selected geography
READY_WITH_RISK (no DR region)	Prod and CVAL placed but no DR region qualifies — check DR free capacity and HC_Gate DR column

Ranked Candidate Table (rows 4+, columns H-P)

The ranked table shows all eligible candidates across the three placement modes (Prod / CVAL / DR) with their placement scores. Use this to understand why a particular region ranked first or fell below the winner.

4. Setting Up a Scenario — Setup Sheet

All scenario parameters are entered in **Setup** column B, rows 3–9. Cells are highlighted yellow. All other cells in Setup are computed and must not be edited.

Cell	Parameter	Input type	Example
B3	Geography	Dropdown: US / EU / Australia / Asia Pacific / Middle East	US
B4	SKU	Dropdown: E16ads_v5 / E32ads_v5 / E8ads_v5 / E4ads_v5 / D8ads_v5 / D4ads_v5 / D2ads_v5 / D16ads_v5	E32ads_v5
B5	VM count (number of VMs for this deployment)	Positive integer	4
B8	Prod region override	Text — Azure region ID; leave blank for auto-selection	eastus2
B9	Customer ID	Text reference	CUST-0042

Computed (do not edit)

Cell	Formula	Meaning
B6	=VLOOKUP(B4,E3:F10,2,FALSE)	vCPU per VM from the SKU catalogue in E3:F10
B7	=B5*B6	Total Requested vCPU — this is the key sizing input used throughout HC_Gate and Scoring
B10	=VLOOKUP(B3,E13:G17,2,FALSE)	Distribution model for the selected geography
B11	=VLOOKUP(B3,E13:G17,3,FALSE)	DR scope geography (US → US; EU/AU/AP → same geo; Middle East → EU for cross-geo DR)

SKU Catalogue (Setup E3:F10)

SKU	vCPU per VM
E32ads_v5	32
E16ads_v5	16
D16ads_v5	16
E8ads_v5	8
D8ads_v5	8
E4ads_v5	4
D4ads_v5	4
D2ads_v5	2

Example: B4 = E32ads_v5 , B5 = 4 → B6 = 32 , B7 = 128 vCPU requested.

5. Updating Capacity Data — Capacity_Usage Sheet

5.1 Which Cells Are Editable

The Capacity_Usage sheet has 22 columns (A–V). Only the columns below accept manual input; the rest are either static catalogue data (locked) or formulas that compute automatically.

Editable columns (yellow):

Col	Header	What to enter
F	Prod Reserved	Total vCPU held in Production capacity reservations for this region
G	Prod Alloc	Total vCPU currently allocated (VM instances consuming Prod reservations)
I	Prod Q-Limit	Production subscription quota limit (vCPU) for the region's VM family
J	Prod Q-Used	Current production quota consumption (vCPU)
L	NP Reserved	Total vCPU in Non-Production (CVAL) capacity reservations
M	NP Alloc	Total vCPU currently allocated in Non-Prod reservations
O	NP Q-Limit	Non-Production subscription quota limit (vCPU)
P	NP Q-Used	Current Non-Prod quota consumption (vCPU)
R	DR Reserved	Total vCPU in DR capacity reservations
S	DR Free	DR vCPU not currently allocated (the available DR pool)
T	DR Coverage	DR coverage ratio (0.0–1.0, e.g. 0.6 = 60% of Prod workload covered)
U	Cust Count	Number of customers currently placed in this region

Computed columns (do not edit — formula cells):

Col	Header	Formula
H	Prod Free	=F-G (Prod Reserved minus Prod Alloc)
K	Prod Q-Headroom	=I-J (Prod Q-Limit minus Prod Q-Used)
N	NP Eff-Free	=L-M (NP Reserved minus NP Alloc)
Q	NP Q-Headroom	=O-P (NP Q-Limit minus NP Q-Used)

Static catalogue columns (do not edit):

Col	Content
A	Region ID (e.g. eastus2)
B	Region Name (e.g. East US 2)
C	Geography (US , EU , Australia , Asia Pacific , Middle East)
D	Distribution Model (3-region , 2-region , cross-geo)
E	Availability Zone count (2 or 3)
V	Class (Standard or Restricted)

Important: Never edit columns A-E or V. These match the ACRME catalogue definition in baseline v2.4 Section 6. Changing them will corrupt HC_Gate and Scoring formulas that reference them by value.

5.2 Worked Example A — Expanding Reserved Capacity

Scenario: Microsoft has provisioned an additional 32,768-core capacity reservation in East US 2 for Production. Update the workbook to reflect the expanded reservation.

Before (East US 2, row 7):

Col	Header	Value
F	Prod Reserved	154,178
G	Prod Alloc	142,756
H	Prod Free (computed)	11,422

Step 1: Click cell **F7** (East US 2, Prod Reserved).

Step 2: Change the value from 154178 to 186946 (154,178 + 32,768).

Step 3: Press Enter. Cell H7 recalculates immediately:

Col	Header	New value
F	Prod Reserved	186,946
G	Prod Alloc	142,756 (unchanged)
H	Prod Free (computed)	44,190

Effect on downstream sheets:

- **HC_Gate row 7 (East US 2):** The Prod quota headroom (K7) drives HC-3 Prod, not Prod Reserved. If the quota limit has not changed, HC-3 Prod stays the same. Prod Free (H) is informational here.
- **Scoring_Prod row 7:** $\alpha_{\text{raw}} = \text{NP Eff-Free} / \text{Prod Reserved} = \text{N7} / \text{F7}$. Increasing F7 slightly lowers α_{raw} — the ratio of NP headroom to prod size decreases. This is by design: a larger regional footprint signals higher demand pressure.

When to update Prod Reserved (F): Whenever Microsoft confirms a new Capacity Reservation Group (CRG) is provisioned or an existing CRG has its count adjusted via the reservation management workflow (CAP-009/CAP-022).

Scenario 2 (within Example A): A batch of 500 E32ads_v5 VMs (= 500 × 32 = 16,000 vCPU) is deployed into East US 2 Production. Update allocation.

Step 1: Click cell **G7** (Prod Alloc).

Step 2: Change from 142756 to 158756 (142,756 + 16,000).

Step 3: Press Enter. H7 recalculates:

Col	Before	After
G	142,756	158,756
H	11,422	−3,810 (deficit!)

The negative H7 value means Prod Alloc exceeds Prod Reserved — a reservation deficit (A.3 from Appendix A). This is **visible but does not trigger a formula error**; it will however cause HC_Gate checks to fail if they rely on Prod Free. Treat any negative H value as an urgent action item: raise a CRG quantity increase.

5.3 Worked Example B — Updating Quota Limits After a Microsoft Increase

Scenario: Microsoft granted a quota increase of 50,000 vCPU for the EDSv5 VM family in West US 3 (Non-Production). Update the workbook.

Before (West US 3, row 4):

Col	Header	Value
O	NP Q-Limit	94,710
P	NP Q-Used	2
Q	NP Q-Headroom (computed)	94,708

Step 1: Click cell **O4** (NP Q-Limit).

Step 2: Change from 94710 to 144710 .

Step 3: Press Enter. Q4 recalculates:

Col	Header	New value
O	NP Q-Limit	144,710
P	NP Q-Used	2 (unchanged)
Q	NP Q-Headroom (computed)	144,708

Effect on downstream sheets:

HC_Gate row 4 re-evaluates:

- **HC-7 DR Integ (I4):** $P4 + \text{Setup!B7} \leq O4 - \text{Policy!B13} \rightarrow 2 + B7 \leq 144,710 - 40$. The expanded limit makes it far easier to PASS HC-7.
- **Scoring_Prod/CVAL/DR row 4:** β_{raw} uses Q / I (Prod) or Q / O (CVAL/DR quota headroom ratio) — increasing O4 or I4 directly improves the β component of the placement score.

When to update quota cells (I, J, O, P): After a Microsoft quota-increase approval is confirmed via the Azure Portal or quota management API. Update both the limit (I/O) and current usage (J/P) from the same observation timestamp to keep the snapshot coherent (DAT-003).

6. Running Multiple-SKU Scenarios

6.1 Why the Tool Models One SKU per Run

The Setup sheet accepts a single SKU (B4) and derives a single Requested vCPU value ($B7 = B5 \times B6$). All HC_Gate checks and placement scores compare against this single vCPU total. The tool does not natively model mixed-SKU deployments in one pass.

This matches the baseline modelling assumption: the HC gate validates that a region can absorb the **largest homogeneous vCPU block** of a customer's deployment in each environment separately.

6.2 Worked Example — E32ads_v5 + E16ads_v5 Dual-SKU Deployment

Scenario: Customer CUST-0099 in the US geography needs:

- **Component A:** 8 × E32ads_v5 VMs = 256 vCPU (workload backbone)
- **Component B:** 4 × E16ads_v5 VMs = 64 vCPU (API layer)

The platform placement decision must accommodate both blocks. Run two separate passes and compare:

Pass 1 — Larger block (E32ads_v5)

Setup cell	Value
B3	US
B4	E32ads_v5
B5	8
B9	CUST-0099

→ B7 = 256 vCPU

→ Note the recommended Prod, CVAL, DR regions from the Result sheet.

→ Example result: **Prod = East US 2, CVAL = West US 3, DR = Central US** (PS = 0.71 / 0.68 / 0.64)

Pass 2 — Smaller block (E16ads_v5)

Setup cell	Value
B3	US
B4	E16ads_v5
B5	4
B9	CUST-0099
B8	eastus2 (← lock Prod to the Pass 1 winner)

→ B7 = 64 vCPU

→ With B8 = eastus2, the engine honours that explicit Prod region override and only competes for CVAL/DR.

→ Check that HC_Gate rows for East US 2, West US 3, and Central US still PASS at 64 vCPU — they should, since $64 < 256$.

Interpreting the results

If **both passes return the same Prod winner** and HC passes at both vCPU sizes, the recommended placement is valid for the combined deployment. Record both requested vCPU values in the customer seed record (PLC-003).

If Pass 1 passes but **Pass 2 fails HC-3 NonProd** in the CVAL region (e.g. West US 3 Q-Headroom < 64 after the 256 vCPU reservation already consumed most of the quota budget), escalate the quota increase for the CVAL subscription before placement.

Summing vCPUs for a conservative gate check

For an even more conservative check, create a third pass using the **combined vCPU total** as a single block:

Setup cell	Value
B4	E32ads_v5 (proxy — pick the larger SKU)
B5	10 (= 8+4, treating all VMs as E32; this overstates the E16 request by 16 vCPU per VM)

This is intentionally conservative and will overfail for very tight scenarios. Use the individual passes as the authoritative check.

Multiple customers with the same dual-SKU profile

If you need to plan capacity for **N customers**, each with the same dual-SKU profile:

1. Run Pass 1 with $B5 = 8 \times N$ to check whether the region can absorb all E32 blocks.
2. Run Pass 2 with $B5 = 4 \times N$ to check E16 blocks.
3. Increment Cust Count (col U) by N for the winning regions after each planning cycle.

7. Calculation Logic Reference

This section documents every formula in the workbook, referenced to the actual Excel cell range. Row variable {r} denotes any data row 4–17 (the 14 ACRME regions).

7.1 Setup Sheet Lookups

vCPU per VM (B6)

```
=VLOOKUP(B4, E3:F10, 2, FALSE)
```

Looks up the selected SKU (B4) in the catalogue table E3:F10 and returns the vCPU count from column F. Returns an error if the SKU is not in the dropdown list.

Requested vCPU (B7)

```
=B5 * B6
```

Multiplies VM count by vCPU per VM. This single value propagates to **every** HC check and scoring formula as Setup!\$B\$7.

Distribution model and DR scope geography (B10, B11)

```
=VLOOKUP(B3, E13:G17, 2, FALSE) ' B10: distribution model
=VLOOKUP(B3, E13:G17, 3, FALSE) ' B11: DR scope geography
```

The table E13:G17 maps each geography to its model and DR scope:

Geography	Distribution Model	DR Scope Geography
US	3-region	US
EU	2-region	EU
Australia	2-region	Australia
Asia Pacific	2-region	Asia Pacific
Middle East	cross-geo	EU

The DR scope geography controls which Scoring_DR regions are InScope (§7.7). Middle East maps to EU, so the DR winner is always a European region (cross-geo DR, DR-020/PLC-010b).

7.2 Capacity_Usage Computed Columns

These four columns update automatically whenever the adjacent editable inputs change.

Prod Free (H)

$$=F\{r\} - G\{r\}$$

Prod Reserved – Prod Alloc = unallocated production reservation capacity. A negative value indicates a **reservation deficit** (Appendix A.3). Does not directly gate eligibility but is used in the builder’s distribution logic and is visible for operational review.

Prod Q-Headroom (K)

$$=I\{r\} - J\{r\}$$

Prod Q-Limit – Prod Q-Used = remaining production quota. This is the **gating value for HC-3 Prod** (§7.3) and the denominator in the β scoring component for Prod (§7.5).

NP Eff-Free (N)

$$=L\{r\} - M\{r\}$$

NP Reserved – NP Alloc = available non-production reservation headroom. This is the **gating value for HC-3 NonProd** first condition and the numerator in the α scoring component (see §7.5–7.6).

NP Q-Headroom (Q)

$$=O\{r\} - P\{r\}$$

NP Q-Limit – NP Q-Used = remaining non-production quota. Used in HC-3 NonProd second condition, HC-7 DR Integrity, and β scoring for CVAL/DR (§7.6–7.7).

7.3 HC_Gate — Hard Constraint Eligibility Checks

The HC_Gate sheet evaluates six binary checks (PASS/FAIL) for each region and derives three eligibility flags (YES/NO). A region must pass **all required checks for an environment** before it enters the scoring stage.

Notation below uses row 7 (East US 2) as the illustrative example.

HC-3 Production Quota (column E)

What it checks: Does this region have enough production quota headroom for the requested deployment, and will a minimum quota buffer remain afterwards?

```
=IF(AND(
    Capacity_Usage!K7 >= Setup!$B$7,
    Capacity_Usage!K7 - Setup!$B$7 >= Policy!$B$15
), "PASS", "FAIL")
```

Component	Value source	Meaning
K7	Prod Q-Headroom = I7 – J7	Current available production quota
Setup!\$B\$7	Requested vCPU	vCPU needed for deployment
Policy!\$B\$15	Min Prod headroom (default: 20 vCPU)	Buffer that must remain after deployment

Condition 1: $K7 \geq B7 \rightarrow$ quota headroom covers the deployment.

Condition 2: $K7 - B7 \geq 20 \rightarrow$ at least 20 vCPU of headroom remains after the deployment. This is the minimum-headroom guardrail (Policy!B15).

Example: Requested = 128 vCPU, East US 2 Prod Q-Headroom = 11,422.

$\rightarrow 11,422 \geq 128 \checkmark$

$\rightarrow 11,422 - 128 = 11,294 \geq 20 \checkmark$

$\rightarrow$ Result: **PASS**

HC-3 NonProd Quota (column F)

What it checks: Does this region have enough non-production reservation headroom for the deployment AND sufficient quota headroom afterwards?

```
=IF(AND(
    Capacity_Usage!N7 >= Setup!$B$7,
    Capacity_Usage!Q7 - Setup!$B$7 >= Policy!$B$16
), "PASS", "FAIL")
```

Component	Value source	Meaning
N7	NP Eff-Free = L7 – M7	Available non-prod reservation capacity
Q7	NP Q-Headroom = O7 – P7	Available non-prod quota
Policy!\$B\$16	Min NonProd headroom (default: 20 vCPU)	Quota buffer required post-deployment

Condition 1: $N7 \geq B7 \rightarrow$ NP reservation headroom covers the deployment.

Condition 2: $Q7 - B7 \geq 20 \rightarrow$ at least 20 vCPU of NP quota remains.

Note: Condition 1 checks reservation headroom (N); condition 2 checks quota headroom (Q). These are independent resources; both must pass (CAP-009/RDY-003).

HC-3 DR Capacity (column G)

What it checks: Does this region hold enough free DR reservation to absorb at least one DR bootstrap event?

```
=IF(Capacity_Usage!S7 >= Policy!$B$13, "PASS", "FAIL")
```

Component	Value source	Meaning
S7	DR Free	Unallocated DR reservation capacity
Policy!\$B\$13	DR bootstrap qty (default: 40 vCPU)	Minimum DR headroom required

Example: DR Free = 16,722 vCPU, bootstrap = 40.

$\rightarrow 16,722 \geq 40 \checkmark \rightarrow$ **PASS**

For the Middle East (Saudi Arabia Central, UAE North), DR Reserved = 0 and DR Free = 0 $\rightarrow$ HC-3 DR always **FAIL** for these regions in the DR scoring sheet. Their DR is served cross-geo in Europe (DR-020).

HC-6 DR Coverage Floor (column H)

What it checks: In a two-region geography where CVAL and DR co-locate (PLC-010a), can the combined NP and DR pool in that region absorb the DR bootstrap demand? This handles the CVAL-sacrifice bootstrap pattern (DR-005/DR-006).

```
=IF(Capacity_Usage!S7 + Capacity_Usage!N7 >= Policy!$B$13, "PASS", "FAIL")
```

Component	Value source	Meaning
S7	DR Free	Unallocated DR capacity
N7	NP Eff-Free	Releasable CVAL capacity
Policy!\$B\$13	DR bootstrap qty (40 vCPU)	Minimum combined DR + releasable CVAL needed

HC-6 recognises that in a co-located CVAL/DR region, CVAL capacity is earmarked as releasable toward DR bootstrap. Even if S7 alone is too low, a region passes HC-6 if $S7 + N7 \geq 40$.

HC-7 DR Floor Integrity (column I)

What it checks: After placing the new deployment in this region's Non-Prod quota, does the remaining NP quota still protect the DR bootstrap floor?

```
=IF(Capacity_Usage!P7 + Setup!$B$7 <= Capacity_Usage!O7 - Policy!$B$13, "PASS", "FAIL")
```

Rearranged: $P7 + B7 \leq O7 - 40$

→ NP Q-Used + Requested vCPU ≤ NP Q-Limit – DR bootstrap

The right-hand side $O7 - 40$ is the NP quota available above the DR bootstrap floor. If adding the new deployment to current NP usage would consume into that floor, the check fails.

Design intent: In a two-region co-located model, the NP subscription's quota must always leave room for the DR minimum — you cannot deploy NP VMs if doing so would crowd out the quota needed to stand up DR bootstrap capacity.

Eligibility Flags (columns J, K, L)

```
' Eligible Prod (J)
=IF(E7="PASS", "YES", "NO")

' Eligible NonProd (K) – requires HC-3 NonProd AND HC-7 integrity
=IF(AND(F7="PASS", I7="PASS"), "YES", "NO")

' Eligible DR (L) – requires HC-3 DR AND HC-6 floor
=IF(AND(G7="PASS", H7="PASS"), "YES", "NO")
```

Flag	Required checks
Eligible Prod	HC-3 Prod only
Eligible NonProd	HC-3 NonProd AND HC-7 DR integrity
Eligible DR	HC-3 DR AND HC-6 DR floor

Eligible NonProd requires HC-7 because adding CVAL load must not break the DR quota floor. Eligible DR requires both HC-3 DR (absolute DR free) and HC-6 (combined pool) to ensure the co-location bootstrap model is safe.

7.4 Scoring Engine — Placement Score (PS)

For each environment (Prod/CVAL/DR), the engine computes a **Placement Score** between 0 and 1 for every eligible region. The score is a weighted sum of five normalised components:

$$PS = \alpha \cdot \alpha_c + \beta \cdot \beta_c + \gamma \cdot \gamma_c + \delta \cdot \delta_c + \epsilon \cdot \epsilon_c$$

Where each `_c` suffix denotes the **clamped** form of the raw component (`MIN(MAX(raw, 0), 1)`), and the weights (α , β , γ , δ , ϵ) are configured in the Policy sheet:

Parameter	Cell	Default	Dimension
α — Capacity headroom weight	Policy!B2	0.30	—
β — Quota headroom weight	Policy!B3	0.20	—
γ — Distribution fairness weight	Policy!B4	0.25	—
δ — DR readiness weight	Policy!B5	0.15	—
ϵ — Zone diversity weight	Policy!B6	0.10	—
Sum	Policy!B7	1.00	Must equal 1.0 (validated in B8)

The weights must sum to 1.0 (within 0.001). Cell B8 shows `VALID` or `INVALID` — always check this after any weight adjustment.

7.5 Scoring_Prod — Formula Details

Scoring_Prod rows 4–17, one row per region. Column letters below refer to Scoring_Prod columns.

InScope (C)

```
=IF(Capacity_Usage!C{r} = Setup!$B$3, 1, 0)
```

1 if the region's geography (Capacity_Usage col C) matches the selected geography (Setup!B3). Only in-scope regions can be Prod candidates.

Eligible (D)

```
=IF(HC_Gate!J{r} = "YES", 1, 0)
```

1 if the region passed Eligible Prod (HC_Gate col J).

AutoSelect (E)

```
=IF(OR(Capacity_Usage!V{r} = "Standard", Capacity_Usage!B{r} = Setup!$B$8), 1, 0)
```

1 if the region is **Standard class** (auto-selectable) OR if its name matches the explicit Prod region override in Setup!B8.

Restricted regions (Saudi Arabia Central, UAE North) have `V = "Restricted"`. They receive `AutoSelect = 0` unless the operator explicitly names them in B8. This enforces the restricted-deployment-region rule from baseline v2.4 Section 6.

α_{raw} (G) — Capacity Headroom

```
=IFERROR(Capacity_Usage!N{r} / Capacity_Usage!F{r}, 0)
```

NP Eff-Free / Prod Reserved

This ratio measures how much non-production reservation capacity (col N) exists relative to the region's total production footprint (col F). A high ratio indicates the region has substantial NP headroom proportional to its Prod size — a proxy for under-utilised capacity available for new customer growth.

Note: This is the Prod scoring α , not a direct prod-headroom ratio. The reasoning is that in a 3-region US geography, the Prod placement decision considers the availability of NP capacity (where CVAL will later land) as a signal of regional capacity health.

α_{c} (H) — Clamped α

```
=MIN(MAX(G{r}, 0), 1)
```

Clamps α_{raw} to [0, 1]. A ratio > 1 is clamped to 1.

β_{raw} (I) — Quota Headroom

```
=IFERROR(Capacity_Usage!K{r} / Capacity_Usage!I{r}, 0)
```

Prod Q-Headroom / Prod Q-Limit

Fraction of production quota still available. A value of 1.0 means no quota is used; 0.0 means quota is exhausted.

Example: East US 2 row 7 — if I7 (Prod Q-Limit) = 200,000 and K7 = 11,422, then $\beta_{\text{raw}} = 11,422 / 200,000 \approx 0.057$. This region is heavily quota-loaded and will score low on β .

β_c (J)

```
=MIN(MAX(I{r}, 0), 1)
```

 γ_{raw} (K) — Distribution Fairness

```
=IFERROR(1 - Capacity_Usage!U{r} / F{r}, 0)
```

```
1 - (Cust Count / total_customers_for_geo)
```

Wait — the formula uses `F{r}` (Prod Reserved) as the denominator, **not** a `total_customers` value. This is actually `1 - (Cust Count / Prod Reserved)`, which approximates how saturated the region is relative to its capacity size.

Design note: The `total_customers` variable (referenced in Policy!B10-B11) is not yet resolved in the baseline (see §9). The current formula uses Prod Reserved as a normalising denominator, which gives a per-vCPU saturation ratio. A region with 150 customers and 154,178 Prod Reserved vCPU (East US 2) scores: `1 - 150/154,178  $\approx$  0.999` — very high, because Prod Reserved dwarfs the customer count. This is a known gap documented in the design.

 γ_c (L)

```
=MIN(MAX(K{r}, 0), 1)
```

 δ_{raw} (M) — DR Readiness

```
=IFERROR(Capacity_Usage!T{r}, 0)
```

`DR Coverage` — directly the value in Capacity_Usage col T (a 0.0–1.0 ratio). A region with 80% of Prod workload covered by DR capacity scores 0.80 on δ .

Example: West US 3 (row 4) DR Coverage = 0.6 $\rightarrow \delta_{\text{raw}} = 0.60 \rightarrow \delta_c = 0.60$.

 δ_c (N)

```
=MIN(MAX(M{r}, 0), 1)
```

 ϵ_{raw} (O) — Zone Diversity

```
=Capacity_Usage!E{r} / 3
```

`AZ count / 3`. Regions with 3 availability zones score 1.0; regions with 2 AZ (Australia Southeast) score 0.667. This rewards regions with full zone diversity for HA.

 ϵ_c (P)

```
=MIN(MAX(O{r}, 0), 1)
```

Placement Score (Q)

```
=Policy!$B$2 * H{r} + Policy!$B$3 * J{r} + Policy!$B$4 * L{r}
+ Policy!$B$5 * N{r} + Policy!$B$6 * P{r}
```

$$PS_Prod = \alpha \cdot \alpha_c + \beta \cdot \beta_c + \gamma \cdot \gamma_c + \delta \cdot \delta_c + \epsilon \cdot \epsilon_c$$

$$= 0.30 \cdot \alpha_c + 0.20 \cdot \beta_c + 0.25 \cdot \gamma_c + 0.15 \cdot \delta_c + 0.10 \cdot \epsilon_c$$

All weights sourced live from Policy!B2:B6 — changing a weight immediately updates all 14 PS values.

Candidate (R)

```
=IF(AND(C{r}=1, D{r}=1, E{r}=1), Q{r}, -1)
```

Returns the PS score only if the region is InScope **AND** Eligible **AND** AutoSelect. Otherwise returns -1 (excluded from ranking).

Rank (S)

```
=IF(R{r}<0, "", COUNTIF($R$4:$R$17,">"&R{r}))+1)
```

Rank among all Candidate regions (blank for ineligible). The winning region has Rank = 1.

Winner Cells

Cell	Content
Scoring_Prod!U3	Region name of the Prod winner (highest PS among all Rank=1 candidates)
Scoring_Prod!U4	Winning Prod PS score

7.6 Scoring_CVAL — Formula Details and Differences

Scoring_CVAL uses the same 19-column structure as Scoring_Prod but differs in three formulas:

α _raw (G) — NP Capacity Headroom

```
=IFERROR(Capacity_Usage!N{r} / Capacity_Usage!L{r}, 0)
```

NP Eff-Free / NP Reserved

For CVAL placement, α measures Non-Prod capacity headroom as a fraction of total NP Reserved — a cleaner utilisation ratio than the Prod version.

β _raw (I) — NP Quota Headroom

```
=IFERROR(Capacity_Usage!Q{r} / Capacity_Usage!O{r}, 0)
```

NP Q-Headroom / NP Q-Limit (Non-Prod quota, not Prod quota).

δ_{raw} (M) — δ duplicates α in CVAL

```
=IFERROR(Capacity_Usage!N{r} / Capacity_Usage!L{r}, 0)
```

This is identical to α_{raw} . This is a **known design gap** — the δ (DR readiness) component for CVAL scoring is not yet defined independently and currently uses the same formula as α . The effective result is that α and δ are perfectly correlated in CVAL scoring, giving the NP capacity headroom dimension a combined weight of $\alpha + \delta = 0.30 + 0.15 = 0.45$. See §9 for the documented gap.

Eligible (D) — uses NonProd flag

```
=IF(HC_Gate!K{r}="YES", 1, 0)
```

Uses HC_Gate column K (Eligible NonProd) rather than column J (Eligible Prod).

Candidate (R) — excludes the Prod winner

```
=IF(AND(C{r}=1, D{r}=1, E{r}=1, Capacity_Usage!B{r}<>Scoring_Prod!$U$3), Q{r}, -1)
```

The additional condition $B \neq \text{Scoring_Prod!U3}$ ensures the Prod winner region is excluded from CVAL candidates (ENV-003 separation — Prod and CVAL should not co-locate unless the distribution model forces it, e.g. two-region geographies per PLC-010a).

7.7 Scoring_DR — Formula Details and Differences

InScope (C) — uses DR scope geography (B11)

```
=IF(Capacity_Usage!C{r} = Setup!$B$11, 1, 0)
```

For US geography, B11 = US → only US regions are in scope for DR.

For Middle East, B11 = EU → **only European regions** are DR candidates (cross-geo DR, DR-020/PLC-010b). This is how the model enforces that Middle East DR always lands in Europe.

α_{raw} (G) — DR Capacity Utilisation

```
=IFERROR(Capacity_Usage!S{r} / Capacity_Usage!R{r}, 0)
```

DR Free / DR Reserved — fraction of DR reservation capacity still available.

β_{raw} (I) — NP Quota Headroom (same as CVAL)

```
=IFERROR(Capacity_Usage!Q{r} / Capacity_Usage!O{r}, 0)
```

Reuses NP quota headroom ratio — the DR region's NP quota is relevant because DR capacity frequently co-locates with CVAL quota (PLC-010a).

δ_raw (M) — DR Coverage vs Target

```
=IFERROR(Capacity_Usage!T{r} / Policy!$B$14, 0)
```

DR Coverage / DR coverage target (0.8) — how close is this region to its DR coverage target? A region at 0.6 coverage with target 0.8 scores $0.6/0.8 = 0.75$. This incentivises choosing regions where DR coverage is already near-sufficient so the new customer's DR portion is well-supported.

Eligible (D) — uses DR flag

```
=IF(HC_Gate!L{r}="YES", 1, 0)
```

Uses HC_Gate column L (Eligible DR) — requires both HC-3 DR and HC-6 DR floor.

Candidate (R) — excludes both Prod and CVAL winners

```
=IF(AND(C{r}=1, D{r}=1, E{r}=1,
        Capacity_Usage!B{r}<>Scoring_Prod!$U$3,
        Capacity_Usage!B{r}<>Scoring_CVAL!$U$2),
    Q{r}, -1)
```

Excludes both the Prod winner and the CVAL winner from DR candidates. In a 3-region geography (US), this enforces full separation of all three environments (ENV-003). In a 2-region geography, the CVAL winner is the only available non-Prod region, so it will be the same as the DR winner — but that is handled at the `AutoSelect` level (the CVAL region has been designated by PLC-010a).

7.8 Result Sheet Derivation

The Result sheet pulls the winner values from each Scoring sheet:

Cell	Formula (approximate)	Meaning
B9	=IF(Scoring_Prod!U3="", "NONE ELIGIBLE", Scoring_Prod!U3)	Prod winner or NONE ELIGIBLE
C9	=Scoring_Prod!U4	Prod winning PS
B10	=IF(Scoring_CVAL!U2="", "NONE ELIGIBLE", Scoring_CVAL!U2)	CVAL winner
C10	=Scoring_CVAL!U3	CVAL winning PS
B11	=IF(Scoring_DR!U2="", "NONE ELIGIBLE", Scoring_DR!U2)	DR winner
C11	=Scoring_DR!U3	DR winning PS
B13	Nested IF on B9/B10/B11	Readiness state

The **Readiness state** formula in B13 evaluates the three winner cells in order:

```
If B9 = "NONE ELIGIBLE" → "QUOTA_DEFICIT / NEEDS EXPLICIT REGION (Prod)"
Else if B10 = "NONE ELIGIBLE" → "CAPACITY_UNAVAILABLE (CVAL)"
Else if B11 = "NONE ELIGIBLE" → "READY_WITH_RISK (no DR region)"
Else → "READY"
```

This maps to the RDY-002 machine-readable states (**READY** , **READY_WITH_RISK** , **QUOTA_DEFICIT** , **CAPACITY_UNAVAILABLE**) from Section 10 of the baseline.

8. Policy Sheet — Tuning Weights and Thresholds

Weight Parameters (B2:B6)

Edit these to reprioritise the placement scoring. After any change, verify **B7 = 1.000** and **B8 = "VALID"** before interpreting results.

Cell	Parameter	Default	Effect of increasing
B2	α — Capacity headroom	0.30	Favours regions with more free NP/DR reservation capacity
B3	β — Quota headroom	0.20	Favours regions with more unused quota margin
B4	γ — Distribution fairness	0.25	Favours regions with fewer existing customers relative to their size
B5	δ — DR readiness	0.15	Favours regions where DR coverage is already strong
B6	ε — Zone diversity	0.10	Favours 3-AZ regions over 2-AZ regions

Example — DR-heavy scenario: To stress-test DR placement when DR coverage is the top priority:

B5 (δ) = 0.40 B2 (α) = 0.20 B3 (β) = 0.15 B4 (γ) = 0.15 B6 (ε) = 0.10
Sum = 1.00 ✓

Headroom Thresholds (B13:B16)

Cell	Parameter	Default	Meaning
B13	DR bootstrap qty vCPU	40	Minimum DR Free required to pass HC-3 DR
B14	DR coverage target	0.80	Target ratio for DR coverage (used in Scoring_DR δ)
B15	Min Prod headroom vCPU	20	Quota buffer required after Prod deployment (HC-3 Prod condition 2)
B16	Min NonProd headroom vCPU	20	Quota buffer required after NonProd deployment (HC-3 NonProd condition 2)

total_customers Mode (B10:B11, D11:E15)

Cell	Parameter	Values
B10	Mode selector	1 = Live / geography sum of Cust Count; 2 = Constant from table; 3 = Manual
B11	Manual total_customers	Numeric — used when B10 = 3
D11:E15	Geography → Constant table	Per-geography constants — used when B10 = 2

Note: In the current workbook, the γ_{raw} formula uses `Prod Reserved` as its denominator rather than `total_customers` (see §9). The B10/B11 mode selector is present in the policy design but is not yet wired into the Scoring sheet γ formula.

9. Known Limitations and Design Gaps

These are **documented**, expected limitations of the v2.4 mockup — not bugs.

1. total_customers Not Wired Into γ_{raw}

Expected design: $\gamma_{\text{raw}} = 1 - (\text{Cust Count} / \text{total_customers_for_geo})$, where `total_customers` is fetched from Policy (B10-B11) or computed as a geography sum.

Current state: γ_{raw} uses `Prod Reserved` as the denominator: $1 - U\{r\}/F\{r\}$. Because `Prod Reserved` ($O(10^5)$ vCPU) is orders of magnitude larger than `Cust Count` ($O(1)$ – $O(150)$), γ_c is always near 1.0 for all regions, making γ effectively a constant and not a discriminating factor in placement.

Impact: Placement decisions are driven by α , β , δ , and ϵ . Distribution fairness is not yet a live differentiator.

Action: Define and agree `total_customers` per geography (Policy!B10-B11 mode), then update the γ_{raw} formula accordingly.

2. δ Duplicates α in Scoring_CVAL

Expected design: δ (DR readiness) and α (capacity headroom) should use independent input metrics.

Current state: In `Scoring_CVAL`, both δ_{raw} and α_{raw} resolve to `NP Eff-Free / NP Reserved`. The δ component in CVAL carries no additional information.

Impact: Effective combined NP-headroom weight in CVAL = $\alpha + \delta = 0.45$ (double-weighted). The distinction between “capacity” and “DR readiness” is lost for CVAL placement.

Action: Define a suitable CVAL DR readiness metric (e.g. `DR Coverage × DR Free` for the co-located CVAL/DR region) and update `Scoring_CVAL!M{r}`.

3. Middle East Capacity Data Is Synthetic

Saudi Arabia Central and UAE North have no real usage data in the source files. The builder injects a synthetic geography pool (`ME MOCK_PROD = 2,400 vCPU`, `ME MOCK_NP = 3,400 vCPU`) and distributes it between the two restricted regions. All Middle East Capacity_Usage values are **derived planning estimates**, not observed consumption.

Action: Replace with observed Azure subscription usage once live customers exist in Middle East regions.

4. Canada Central Data Is Absent — Mock Subscriber Count Used

Canada Central does not appear in the source usage file. The builder assigns a mock subscriber count (10) and distributes a synthetic floor share from the US geography total. Actual usage may differ significantly.

5. Reserved Capacity and Quota Limits Are Derived Planning Figures

The `Prod Reserved`, `NP Reserved`, `DR Reserved`, `Prod Q-Limit`, and `NP Q-Limit` values are **computed from the usage distribution algorithm**, not read from Azure Capacity Reservation APIs or Azure Quota APIs. They represent what should be reserved/requested given the modelled distribution — not what is provisioned today.

Action: Reconcile these planning figures against actual CRG quantities (CAP-009) and quota assignments after each quarterly planning cycle.

6. Japan East Excluded

Japan East is noted as pending confirmation in baseline v2.4 Section 6 and is not in the catalogue. When Japan East is added to the baseline region list, the builder must be updated and the workbook rebuilt.

10. Quick Reference Tables

Capacity_Usage — What to Update and When

Event	Column(s) to update	Formula that re-computes
New CRG provisioned (Prod)	F (Prod Reserved) ↑	H (Prod Free)
New CRG provisioned (Non-Prod)	L (NP Reserved) ↑	N (NP Eff-Free)
New CRG provisioned (DR)	R (DR Reserved) ↑	—
VMs deployed to Prod	G (Prod Alloc) ↑	H (Prod Free)
VMs deployed to NonProd	M (NP Alloc) ↑	N (NP Eff-Free)
DR VMs allocated	S (DR Free) ↓	—
Quota increase granted (Prod)	I (Prod Q-Limit) ↑	K (Prod Q-Headroom)
Quota increase granted (Non-Prod)	O (NP Q-Limit) ↑	Q (NP Q-Headroom)
Quota consumed (Prod deployment)	J (Prod Q-Used) ↑	K (Prod Q-Headroom)
Quota consumed (NonProd deployment)	P (NP Q-Used) ↑	Q (NP Q-Headroom)
DR coverage changed	T (DR Coverage)	—
Customer placed	U (Cust Count) ↑	—

HC_Gate — Pass/Fail Decision Matrix

Check	Pass condition	Environment gated
HC-3 Prod	Prod Q-Headroom $\geq$ Requested vCPU AND remaining $\geq$ 20	Prod eligibility
HC-3 NonProd	NP Eff-Free $\geq$ Requested vCPU AND NP Q-Headroom post-deploy $\geq$ 20	NonProd eligibility
HC-3 DR	DR Free $\geq$ 40 vCPU	DR eligibility (first gate)
HC-6 DR Floor	DR Free + NP Eff-Free $\geq$ 40 vCPU	DR eligibility (second gate)
HC-7 DR Integ	NP Q-Used + Requested vCPU $\leq$ NP Q-Limit – 40	NonProd eligibility (second gate)
Eligible Prod	HC-3 Prod = PASS	Prod enters scoring
Eligible NonProd	HC-3 NonProd = PASS AND HC-7 = PASS	CVAL enters scoring
Eligible DR	HC-3 DR = PASS AND HC-6 = PASS	DR enters scoring

Scoring Component Summary

Component	Prod formula	CVAL formula	DR formula
α_{raw}	NP Eff-Free / Prod Reserved	NP Eff-Free / NP Reserved	DR Free / DR Reserved
β_{raw}	Prod Q-Headroom / Prod Q-Limit	NP Q-Headroom / NP Q-Limit	NP Q-Headroom / NP Q-Limit
γ_{raw}	1 – Cust Count / Prod Reserved	(same formula)	(same formula)
δ_{raw}	DR Coverage (T) directly	NP Eff-Free / NP Reserved (= α, gap)	DR Coverage / DR target (B14)
ϵ_{raw}	AZ count / 3	AZ count / 3	AZ count / 3

Readiness State Cheat Sheet

Result!B13 state	Root cause	Fix
READY	All three placements found	No action
QUOTA_DEFICIT / NEEDS EXPLICIT REGION (Prod)	No Prod candidate: all regions fail HC-3 Prod or AutoSelect=0	1. Increase Prod Q-Limit (I) in target region, or 2. Set Setup! B8 to a Restricted region ID
CAPACITY_UNAVAILABLE (CVAL)	No CVAL candidate in the geography	Increase NP Eff-Free (L–M) or NP Q-Limit (O) in target NonProd region
READY_WITH_RISK (no DR region)	No DR candidate	Increase DR Free (S) or expand DR Reserved (R) in DR-scope geography